Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The system of equations $\begin{pmatrix} 3 & -2 & 1 \\ 5 & -8 & 9 \\ 2 & 1 & a \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} b \\ 3 \\ -1 \end{pmatrix}$ has no solution if a and b are

(A) $a = -3, b \neq 1/3$

(B) $a = 2/3, b \neq 1/3$

(C) $a \neq 1/4, b = 1/3$

(D) $a \neq -3, b \neq 1/3$

Q.2

Let Aand B be two matrices of order $\mathrm{n} \times \mathrm{n}$. Let A be non-singular and B be singular. Consider the following:
1 AB is singular
2 AB is non-singular
3 $\mathrm{A}^{-1}\ \mathrm{B}$ is singular
4 $\mathrm{A}^{-1}\ \mathrm{B}$ is non singular Which of the above is/ are correct?

(A) 1 and 3

(B) 2 and 4 only

(C) 1 only

(D) 3 only

Q.3

If $|A| = 8$, where A is square matrix of order 3, then what is $|\operatorname{adj} A|$ equal to?

(A) 16

(B) 24

(C) 64

(D) 512

Q.4

If the matrix B is the adjoint of the square matrix A and α is the value of the determinant of A, then what is AB equal to ?

(A) α

(B) $\left(\frac{1}{\alpha}\right) \mathrm{I}$

(C) I

(D) $\alpha \mathrm{I}$

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Q.5

If $A = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}$, then what is $A(\text{adj } A)$ equal to

(A) $\begin{bmatrix} 0 & 10 \\ 10 & 0 \end{bmatrix}$

(B) $\begin{bmatrix} 10 & 0 \\ 0 & 10 \end{bmatrix}$

(C) $\begin{bmatrix} 1 & 10 \\ 10 & 1 \end{bmatrix}$

(D) $\begin{bmatrix} 10 & 1 \\ 1 & 10 \end{bmatrix}$

Q.6

If$\text{adj } B = A, |P| = |Q| = 1$, then adj $\left(Q^{-1}BP^{-1}\right)$ is

(A) PQ

(B) QAP

(C) PAQ

(D) $\mathrm{PA^{-1}Q}$

Q.7

If A is a square matrix such that $(\mathrm{A} - 2\mathrm{I})(\mathrm{A} + \mathrm{I}) = \mathrm{O}$, then $\mathrm{A}^{-1} =$

(A) $\frac{\mathrm{A}-\mathrm{I}}{2}$

(B) $\frac{\mathrm{A}+\mathrm{I}}{2}$

(C) $2(\mathrm{A} - \mathrm{I})$

(D) $2\,\mathrm{A} + \mathrm{I}$

Q.8

If A and B are two square matrices such that $\mathrm{B} = -\mathrm{A}^{-1}\mathrm{BA}$, then $(\mathrm{A} + \mathrm{B})^2 =$

(A) O

(B) $\mathrm{A}^2 + \mathrm{B}^2$

(C) $A^2 + 2AB + B^2$

(D) $\mathrm{A} + \mathrm{B}$

Q.9

The matrix $\mathrm{A} = \begin{bmatrix} 1 & 3 & 2 \\ 1 & x-1 & 1 \\ 2 & 7 & x-3 \end{bmatrix}$ will have inverse for every real number x except for

(A) $x = \frac{11 \pm \sqrt{5}}{2}$

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(B) $x = \frac{9 \pm \sqrt{5}}{2}$

(C) $x = \frac{11 \pm \sqrt{3}}{2}$

(D) $x = \frac{9 \pm \sqrt{3}}{2}$

Q.10

If $\mathrm{A} = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & a & 1 \end{bmatrix}$ and $\mathrm{A}^{-1} = \begin{bmatrix} 1/2 & -1/2 & 1/2 \\ -4 & 3 & c \\ 5/2 & -3/2 & 1/2 \end{bmatrix}$ then the value of $\mathrm{a} + \mathrm{c}$ is equal to

(A) 1

(B) 0

(C) 2

(D) none of these

Q.11

If A is any 2×2 matrix such that $\begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \mathrm{A} = \begin{bmatrix} -1 & 0 \\ 6 & 3 \end{bmatrix}$, then what is A equal to?

(A) $\begin{bmatrix} -5 & 1 \\ -2 & 2 \end{bmatrix}$

(B) $\begin{bmatrix} -5 & -2 \\ 1 & 2 \end{bmatrix}$

(C) $\begin{bmatrix} -5 & -2 \\ 2 & 1 \end{bmatrix}$

(D) $\begin{bmatrix} 5 & 2 \\ -2 & -1 \end{bmatrix}$

Q.12

If $P = \begin{bmatrix} \cos(\pi/6) & \sin(\pi/6) \\ -\sin(\pi/6) & \cos(\pi/6) \end{bmatrix}$, $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $\mathrm{Q} = \mathrm{PAP}'$ then $\mathrm{P}'\mathrm{Q}^{2007}\mathrm{P}$ is equal to

(A) $\begin{bmatrix} 1 & 2007 \\ 0 & 1 \end{bmatrix}$

(B) $\begin{bmatrix} 1 & \sqrt{3}/2 \\ 0 & 2007 \end{bmatrix}$

(C) $\begin{bmatrix} \sqrt{3}/2 & 2007 \\ 0 & 1 \end{bmatrix}$

(D) $\begin{bmatrix} \sqrt{3}/2 & -1/2 \\ 1 & 2007 \end{bmatrix}$

Q.13

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Let $A = \begin{pmatrix} 1 & -1 & 1 \\ 2 & 1 & -3 \\ 1 & 1 & 1 \end{pmatrix}$. and 10 $B = \begin{pmatrix} 4 & 2 & 2 \\ -5 & 0 & \alpha \\ 1 & -2 & 3 \end{pmatrix}$. If B is the inverse of matrix A, then α is

(A) 5

(B) -1

(C) 2

(D) -2

Q.14

If a matrix A is such that $3A^3 + 2A^2 + 5A + I = 0$
Then what is A^{-1} equal to?

(A) $-\left(3A^2 + 2A + 5I\right)$

(B) $3A^2 + 2A + 5I$

(C) $3A^2 - 2A - 5I$

(D) $\left(3A^2 + 2A - 5I\right)$

Q.15

$A = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 3 \\ -1 & -2 \end{bmatrix}$, then which of the following is/are correct?

1. $\mathrm{AB}\left(\mathrm{A}^{-1}\,\mathrm{B}^{-1}\right)$ is a unit matrix.
2. $(\mathrm{AB})^{-1} = \mathrm{A}^{-1}\,\mathrm{B}^{-1}$

Select the correct answer using the code given below:

(A) 1 only

(B) 2 only

(C) Both 1 only 2

(D) Neither 1 nor 2

Q.16

Inverse of the matrix $\begin{bmatrix} 3 & -2 & -1 \\ -4 & 1 & -1 \\ 2 & 0 & 1 \end{bmatrix}$ is

(A) $\begin{bmatrix} 1 & 2 & 3 \\ 3 & 3 & 7 \\ -2 & -4 & -5 \end{bmatrix}$

(B) $\begin{bmatrix} 1 & -3 & 5 \\ 7 & 4 & 6 \\ 4 & 2 & 7 \end{bmatrix}$

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(C) $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 7 \\ -2 & -4 & -5 \end{bmatrix}$

(D) $\begin{bmatrix} 1 & -3 & 5 \\ 7 & 4 & 6 \\ 4 & 2 & -7 \end{bmatrix}$

Q.17

If k is a scalar and I is a unit matrix of order 3 , then $\mathrm{adj}(kI) =$

(A) $k^3 I$

(B) $k^2 I$

(C) $-k^3 I$

(D) $-k^2 I$

Q.18

If $A^2 - A + I = 0$, then $A^{-1} =$

(A) A^{-2}

(B) $A + I$

(C) $I - A$

(D) $A - I$

Q.19

If $A = \begin{bmatrix} -1 & 2 \\ 2 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$, $AX = B$, then $X =$

(A) [5 7]

(B) $\frac{1}{3}\begin{bmatrix} 5 \\ 7 \end{bmatrix}$

(C) $\frac{1}{3}[57]$

(D) $\begin{bmatrix} 5 \\ 7 \end{bmatrix}$

Q.20

A, B are n-rowed square matrices such that $AB = O$ and B is non-singular. Then

(A) $A \neq O$

(B) $A = O$

(C) $A = I$

(D) None of these

Answers & Solutions

Q.1 Answer:

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$a=-3, b \neq 1/3$

Solution:

The augmented matrix is given by

$$(A \mid B)=\left(\begin{array}{ccc|c} 3 & -2 & 1 & b \\ 5 & -8 & 9 & 3 \\ 2 & 1 & a & -1 \end{array}\right)$$

Applying $R_1 \rightarrow 2R_1 - R_2$**, we get**

$$(A \mid B) \sim \left(\begin{array}{ccc|c} 1 & 4 & -7 & 2b-3 \\ 5 & -8 & 9 & 3 \\ 2 & 1 & a & -1 \end{array}\right)$$

Applying $R_2 \rightarrow R_2 - 5R_1, R_3 \rightarrow R_3 - 2R_1$**, we get**

$$(A \mid B) \sim \left(\begin{array}{ccc|c} 1 & 4 & -7 & 2b-3 \\ 0 & -28 & 44 & 18-10b \\ 0 & -7 & a+14 & 5-4b \end{array}\right)$$

The system of equations will have no solution if

$\frac{-28}{-7}=\frac{44}{a+14} \neq \frac{18-10b}{5-4b}$

$\Rightarrow a+14=11$ **and** $20-16b \neq 18-10b$

$\Rightarrow a=-3$ **and** $b \neq -1/3$

Q.2 Answer:

1 and 3

Solution:

Since, |B|=0

AB and $A^{(-1)}B$ **is singular ie their det=0**

Q.3 Answer:

64

Solution:

Let $|\mathrm{A}|=8$ **and** A **is a square matrix of order 3 . We know that** $|\operatorname{adj} \mathrm{A}|=|\mathrm{A}|^{n-1}$**. I where 'n' is the order of the matrix** A**.**

$\therefore \quad |\operatorname{adj} \mathrm{A}|=8^{3-1}=8^2=64$

Q.4 Answer:

$\alpha \mathrm{I}$

Solution:

Since, adjoint of the square matrix A is B $\Rightarrow \frac{\mathrm{B}}{|\mathrm{A}|}=\mathrm{A}^{-1}$

$\Rightarrow \frac{\mathrm{AB}}{|\mathrm{A}|}=\mathrm{AA}^{-1}=\mathrm{I}$

$\Rightarrow \mathrm{AB}=|\mathrm{A}| \mathrm{I} \Rightarrow \mathrm{AB}=\alpha \mathrm{I}$

Q.5 Answer:

$$\left[\begin{array}{cc} 10 & 0 \\ 0 & 10 \end{array}\right]$$

Solution:

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Let $A = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}$

We have

If A is a square matrix of order n then $A(\text{adj } A) = |A|.\text{I}_\text{n}$

Here, $n = 2$

$\therefore \quad A(\text{adj } A) = \text{I}_2|A|$

$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{vmatrix} 3 & 2 \\ 1 & 4 \end{vmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} (12 - 2) = 10 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$= \begin{bmatrix} 10 & 0 \\ 0 & 10 \end{bmatrix}$

Q.6 Answer:

PAQ

Solution:

Given adj $\text{B} = \text{A}, |\text{P}| = |\text{Q}| = 1$

Consider, $\text{adj}\left(\text{Q}^{-1}\,\text{BP}^{-1}\right)$

$= \left(\text{adj}\,\text{P}^{-1}\right)\left(\text{adj}\,\text{B}\right)\left(\text{adj}\,\text{Q}^{-1}\right)$

$= \left(\text{adj}\,\text{P}\right)^{-1}\,\text{A}\cdot\left(\text{adj}\,\text{Q}\right)^{-1}$

$= \left(\text{P}^{-1}\right)^{-1}\,\text{A}\left(\text{Q}^{-1}\right)^{-1}\left(\because \text{P}^{-1} = \frac{1}{|\text{P}|}\cdot\text{adj}\,\text{P}\right)$

$= \text{PAQ}.$

Q.7 Answer:

$\frac{\text{A}-\text{I}}{2}$

Solution:

$(\text{A} - 2\text{I})(\text{A} + \text{I}) = 0$

$\Rightarrow \text{AA} - \text{A} - 2\text{I} = 0 \quad (\because \text{AI} = \text{A})$

$\Rightarrow A\left(\frac{A-I}{2}\right) = I \quad \therefore \frac{A-I}{2} = A^{-1}$

Q.8 Answer:

$\text{A}^2 + \text{B}^2$

Solution:

$\text{B} = -\text{A}^{-1}\text{BA} \Rightarrow \text{AB} = -\text{BA} \Rightarrow \text{AB} + \text{BA} = 0$

$\therefore (\text{A} + \text{B})^2 = \text{A}^2 + \text{AB} + \text{BA} + \text{B}^2 = \text{A}^2 + \text{B}^2$

Q.9 Answer:

$x = \frac{11 \pm \sqrt{5}}{2}$

Solution:

$\text{A} = \begin{bmatrix} 1 & 3 & 2 \\ 1 & \text{x} - 1 & 1 \\ 2 & 7 & \text{x} - 3 \end{bmatrix}$

$|\text{A}| = 1[(\text{x} - 1)(\text{x} - 3) - 7] - 3[(\text{x} - 3) - 2]$

$+2[7 - 2(x - 1)]$

$= x^2 - 11x + 29$

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If inverse will not exist then $|A| = 0\ x^2 - 11x + 29 = 0$

$x = \frac{11 \pm \sqrt{5}}{2}$

Q.10 Answer:

0

Solution:

We have, $I = AA^{-1}$

$$= \frac{1}{2}\begin{bmatrix} 0 & 1 & 3 \\ 1 & 2 & 3 \\ 3 & a & 1 \end{bmatrix}\begin{bmatrix} 1 & -1 & 1 \\ -8 & 6 & 2c \\ 5 & -3 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & c+1 \\ 0 & 1 & 2(c+1) \\ 4(1-a) & 3(a-1) & 2+ac \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Comparing the elements we get $c + 1 = 0 \Rightarrow c = -1$ **and** $a - 1 = 0 \quad \Rightarrow a = 1$

Q.11 Answer:

$$\begin{bmatrix} -5 & -2 \\ 2 & 1 \end{bmatrix}$$

Solution:

Let $\begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} = B$

Then $BA = \begin{bmatrix} -1 & 0 \\ 6 & 3 \end{bmatrix}$

$$\Rightarrow \quad A = B^{-1}\begin{bmatrix} -1 & 0 \\ -6 & 3 \end{bmatrix}$$

$|B| = 3,$

$$\text{adj}\, B = \begin{bmatrix} 3 & -2 \\ 0 & 1 \end{bmatrix}$$

$$B^{-1} = \frac{1}{3}\begin{bmatrix} 3 & -2 \\ 0 & 1 \end{bmatrix}$$

$\Rightarrow \quad A =$

$$= \begin{bmatrix} -5 & -2 \\ 2 & 1 \end{bmatrix}$$

Q.12 Answer:

$$\begin{bmatrix} 1 & 2007 \\ 0 & 1 \end{bmatrix}$$

Solution:

Note that $P' = P^{-1} \quad$ **Now,** $Q = PAP' = PAP^{-1}$

$\Rightarrow \quad Q^{2007} = PA^{2007}P^{-1}$

$\therefore \quad P'Q^{2007}P = P^{-1}\left(PA^{2007}P^{-1}\right)P$

$= A^{2007} = (I + B)^{2007}$

where $B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$

As $B^2 = 0$**, we get** $B^r = 0 \forall r \geq 2.$

Thus, by binomial theorem, $A^{2007} = I + 2007\ B = \begin{bmatrix} 1 & 2007 \\ 0 & 1 \end{bmatrix}$

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Q.13 Answer:

5

Solution:

Here, $\Rightarrow B=\frac{1}{10}\begin{bmatrix}4 & 2 & 2\\ -5 & 0 & \alpha\\ 1 & -2 & 3\end{bmatrix}$

Also since, $B=A^{-1}\Rightarrow AB=I$

$\Rightarrow \frac{1}{10}\begin{bmatrix}1 & -1 & 1\\ 2 & 1 & -3\\ 1 & 1 & 1\end{bmatrix}\begin{bmatrix}4 & 2 & 2\\ -5 & 0 & \alpha\\ 1 & -2 & 3\end{bmatrix}=\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}$

$\Rightarrow \frac{1}{10}\begin{bmatrix}10 & 0 & 5-2\\ 0 & 10 & -5+\alpha\\ 0 & 0 & 5+\alpha\end{bmatrix}=\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}$

$\Rightarrow \frac{5-\alpha}{10}=0\Rightarrow \alpha=5$

Q.14 Answer:

$-\left(3A^2+2A+5I\right)$

Solution:

Let A be a matrix such that $3\,\mathrm{A}^3+2\,\mathrm{A}^2+5\,\mathrm{A}+\mathrm{I}=0$

Post multiply by A^{-1} **on both the sides, we get** $3\,\mathrm{A}^3\,\mathrm{A}^{-1}+2\,\mathrm{A}^2\,\mathrm{A}^{-1}+5\mathrm{A}\mathrm{A}^{-1}+\mathrm{I}\mathrm{A}^{-1}=0$

$\Rightarrow 3\,\mathrm{A}^2+2\,\mathrm{A}+5\mathrm{I}+\mathrm{A}^{-1}=0$

$\Rightarrow \quad A^{-1}=-\left(3A^2+2A+5I\right)$

Q.15 Answer:

Neither 1 nor 2

Solution:

Here, $A=\begin{bmatrix}1 & -1\\ 2 & 3\end{bmatrix}$ **and** $B=\begin{bmatrix}2 & 3\\ -1 & -2\end{bmatrix}$

$A\mid=3-(-2)=5$ **and** $|B|=-4-(-3)=-1$

$\Rightarrow A^{-1}=\frac{1}{5}\begin{bmatrix}3 & 1\\ -2 & 1\end{bmatrix}$ **and** $B^{-1}=-1\begin{bmatrix}-2 & -3\\ 1 & 2\end{bmatrix}$

$AB=\begin{bmatrix}3 & 5\\ 1 & 0\end{bmatrix}$ **and** $A^{-1}B^{-1}=\frac{1}{5}\begin{bmatrix}5 & 7\\ -5 & 8\end{bmatrix}$

$\Rightarrow AB\left(A^{-1}B^{-1}\right)=\frac{1}{5}\begin{bmatrix}-10 & -12\\ 5 & 7\end{bmatrix}\neq 1$

$AB\mid=0-5=-5$

$\therefore (AB)^{-1}=\frac{-1}{5}\begin{bmatrix}0 & -5\\ -1 & 3\end{bmatrix}\neq A^{-1}B^{-1}$

Q.16 Answer:

$\begin{bmatrix}1 & 2 & 3\\ 2 & 5 & 7\\ -2 & -4 & -5\end{bmatrix}$

Solution:

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Let $A=\begin{bmatrix} 3 & -2 & -1 \\ -4 & 1 & -1 \\ 2 & 0 & 1 \end{bmatrix}$, then $|A|=\begin{vmatrix} 3 & -2 & -1 \\ -4 & 1 & -1 \\ 2 & 0 & 1 \end{vmatrix}=1$ The matrix of cofactors of $A=\begin{bmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{bmatrix}=\begin{bmatrix} 1 & 2 & -2 \\ 2 & 5 & -4 \\ 3 & 7 & -5 \end{bmatrix}$ Therefore, $adj(A)=\begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 7 \\ -2 & -4 & -5 \end{bmatrix}\therefore$ $A^{-1}=\frac{1}{|A|}\cdot adjA=\begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 7 \\ -2 & -4 & -5 \end{bmatrix}, \quad (\because |A|=1).$

Q.17 Answer:

k^2I

Solution:

Let $I=\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$**, then** $kI=\begin{bmatrix} k & 0 & 0 \\ 0 & k & 0 \\ 0 & 0 & k \end{bmatrix}\Rightarrow adj(kI)=\begin{bmatrix} k^2 & 0 & 0 \\ 0 & k^2 & 0 \\ 0 & 0 & k^2 \end{bmatrix}=k^2I$

Q.18 Answer:

$I-A$

Solution:

$A^2-A+I=0\Rightarrow I=A-A^2\Rightarrow I=A(I-A)\Rightarrow A^{-1}I=A^{-1}(A(I-A))\Rightarrow A^{-1}=I-A$

Q.19 Answer:

$\frac{1}{3}\begin{bmatrix} 5 \\ 7 \end{bmatrix}$

Solution:

$A=\begin{bmatrix} -1 & 2 \\ 2 & -1 \end{bmatrix}, B=\begin{bmatrix} 3 \\ 1 \end{bmatrix}\ AX=B\Rightarrow X=A^{-1}BA^{-1}=\frac{adjA}{|A|}\Rightarrow A^{-1}=\frac{-1}{3}\begin{bmatrix} -1 & -2 \\ -2 & -1 \end{bmatrix}$ $=\frac{1}{3}\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ **and** $X=A^{-1}B=\begin{bmatrix} \frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} \end{bmatrix}\begin{bmatrix} 3 \\ 1 \end{bmatrix};X=\frac{1}{3}\begin{bmatrix} 5 \\ 7 \end{bmatrix}$

Q.20 Answer:

$A=O$

Solution:

Since $|B|\neq 0\Rightarrow B^{-1}$ **exists,** $AB=0\Rightarrow (AB)B^{-1}=OB^{-1}\Rightarrow A\left(BB^{-1}\right)=O\Rightarrow AI=O\Rightarrow A=O$

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